

Gamma knife radiosurgery for uveal melanoma: 12 years of experience

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Presented in part at the ARVO annual meeting 2006.

Accepted 6 August 2008
Published Online First
29 August 2008

ABSTRACT

Aim: To present our treatment protocol and evaluate the results of Gamma knife radiosurgery (GKR) in treating patients with uveal melanoma.

Methods: Seventy-eight consecutive patients with uveal melanoma underwent stereotactic radiosurgery (radiation dose 30–50 Gy) with a Leksell Gamma-Knife at the San Raffaele University Hospital, Milan, Italy between 1994 and 2006. The main outcome measures evaluated were: survival rate, local tumour control, eye retention rate, visual acuity and treatment-related complications.

Results: Survival rate was 88.8% at 3 years and 81.9% at 5 years. Local tumour control was achieved in 91.0% of patients. The median tumour thickness reduction after treatment was 1.96 mm ($p < 0.0001$) (–32.1%). The eye retention rate was 89.7%. A significant relative reduction of visual acuity was observed during follow-up. The most frequent treatment-related complications were: exudative retinopathy (33.3%), neovascular glaucoma (18.7%), radiogenic retinopathy (13.5%) and vitreous haemorrhages (10.4%).

Conclusion: GKR can be considered an alternative to enucleation for the treatment of choroidal melanomas.

Uveal melanoma represents the most common form of primary intraocular malignancy in adults. The estimated incidence of this disease in the United States is 4.9 cases per million per year,¹ with 2.8 cases per million per year in Europe.²

Up to the late 1970s, enucleation was traditionally used to treat uveal melanoma, until Zimmerman *et al* proposed the hypothesis that enucleation could increase metastatic disease.³ In recent years, this assumption has led to the development of several conservative treatment strategies.^{4–7}

Gamma knife radiosurgery (GKR) was initially introduced to successfully treat intracranial lesions such as brain tumours, vascular abnormalities, skull base tumours and neurological functional diseases.⁸ Gamma knife radiosurgery has been shown to be an alternative to enucleation for the treatment of large uveal melanomas.^{9–10}

The purpose of the present study was to describe our treatment protocol and evaluate the results of GKR on 78 patients with uveal melanoma treated at the San Raffaele Hospital, Milan (Italy) between 1994 and 2006.

MATERIAL AND METHODS

Patients

Records of 78 consecutive patients with uveal melanoma treated with gamma knife radiosurgery from June 1994 to December 2006 at the

Ophthalmology Department of the San Raffaele Hospital, Milan were retrospectively reviewed.

The diagnosis and clinical characteristics of uveal melanoma were established on the basis of complete ophthalmological evaluation, A- and B-scan ultrasonography, fluorescein and indocyanine green angiography. In adherence with the Declaration of Helsinki, every patient supplied written informed consent before entering the study. Patients were also aware of all current treatment methods for uveal melanoma.

Inclusion criteria were: patients affected by previously untreated uveal melanoma, tumour thickness ≥ 3 mm, patients eligible to be examined with brain MRI without anatomical or functional impairment, complete and precise data collection regarding mortalities. Our patients treated with GKR were not eligible for other conservative treatment modalities currently available in Italy.

Methods

The treatment protocol was performed as follows.

Extraocular muscle sutures

After the patient received retrobulbar anesthesia with long-acting agents (5 cm³ of 1% Ropivacaine) to obtain complete akinesia, two extraocular muscles were sutured through the conjunctiva using 3.0 black silk suture. The two muscles were chosen according to tumour location to optimise globe position during treatment.

Stereotactic lightweight aluminium frame fixation

The stereotactic frame was attached to the patient's head with four pins lodged in the outer plate of the skull. The frame provided the coordinate system for target determination by magnetic resonance imaging.

Globe immobilisation and orientation

The threads of the two sutured muscles were fixed to the stereotactic frame to immobilise and orientate the globe. The globe is oriented in order to localise the tumour as closely as possible to the centre of the stereotactic frame. This condition may reduce magnetic resonance image distortion. Correct globe immobilisation was crucial to performing precise GKR.

Neuroimaging

High-resolution magnetic resonance (2 mm slices) with gadolinium of the brain was performed, and the images were transferred to the gamma knife three-dimensional treatment planning system (Gamma Plan). The use of gadolinium increased

definition of tumour margins in the presence of subretinal fluid and retinal detachment.

Dose planning

The melanoma margins were recognised on magnetic resonance image sections in three-dimensional reconstructions (axial, sagittal and coronal) and drawn on a computer screen with an average 2 mm exceeding margin around tumour edges. We used the "conformation technique" that allowed us to create a field of irradiation corresponding to the tumour shape and spare healthy peritumorous tissue to reduce subsequent side effects. The maximum dose was delivered to the tumour at 100% isodose line. The periphery of the tumour was contained in the 50% isodose curve. Seven patients received 50 Gy at 50% isodose (1994 through 1995), 21 patients received 40 Gy at 50% isodose (1995 through 1999), and 47 patients received 35 Gy at 50% isodose (2000 through 2006). The size of collimators, number of shots, and selective plugging of sources were chosen to optimise the "dose conformation" and spare the optic nerve and the fellow eye (fig 1).

Leksell Gamma Knife

The mechanical precision of our GK is ± 0.3 mm. The high-resolution magnetic resonance has a voxel size of 1.5 mm^3 . The combined system precision (imaging and treatment) is ± 0.5 mm.

Treatment

The patient was positioned prone or supine in the stereotactic attachment inside the selected collimator helmet using the predetermined tumour coordinates. The affected globe, fixed to the frame, was immobilised during treatment. Treatment was performed in a single session, and the whole procedure lasted from 3 to 4 h.

After GKR was carried out, the stereotactic frame and the sutures were removed, and the patients were discharged the following day having been prescribed local antibiotic and anti-inflammatory treatment.

Follow-up

All patients underwent a complete ophthalmological examination including measurement of visual acuity (Snellen charts) 1 day after GKR and then at 1, 3 and 6 months; follow-up was thereafter carried out at 6-month intervals. Colour fundus photos, A- and B-scan ultrasonography were performed at every control visit. Liver ultrasonography and liver function test analyses were regularly repeated every 6 months in order to detect early liver metastases.

Statistical analysis

Results of continuous variables are expressed as medians (interquartile ratio (IQR)) or frequencies (%) for qualitative variables. Absolute and relative variations were calculated for the tumour thickness and visual acuity.

The Wilcoxon sign rank test for paired data was used to compare distribution of continuous variables. The χ^2 test was applied to compare qualitative distribution. The Cochran–Armitage test was calculated to assess a linear trend in proportions along ordinal variables.

The linear correlation between continuous variables was assessed by the Spearman correlation coefficient.

The time to death was estimated using the Kaplan–Meier product-limit method. Survival curves were calculated according

to strata of the baseline factors and compared using the logrank test. Strata of the continuous variables were determined on median values.

All *p* values were two-tailed and were considered as significant if they were <0.05 . The analysis was performed using SAS software (version 8.02; SAS Institute, Cary, North Carolina).

RESULTS

The study population was made up of 78 patients (37 men and 41 women) with a median age at time of diagnosis of 64 years (IQR 58 to 71). The median follow-up time was 31.3 months (IQR 17.6 to 60.6). Patient characteristics are summarised in table 1. The tumour was localised in 37 right eyes and in 41 left eyes with a median tumour thickness of 6.1 mm (IQR 4.7 to 8.8). Data concerning tumour location are summarised in table 1.

Survival rate

Eleven patients died during follow-up. Three patients died of causes unrelated to the tumour, and eight patients died of metastatic disease. According to these data, the 3-year survival rate was 88.8 (SE 11.2)% and the 5-year survival rate was 81.9 (18.1)%. The survival rate is shown in fig 2.

The survival rate was also evaluated in relation to baseline factors that may influence death such as age, sex, tumour location, laterality, Gy at tumour margin and tumour thickness (table 2).

The survival rate was also evaluated in relation to follow-up data: metastasis, recurrence and relative tumour thickness variation (table 3). There was a strong negative statistical correlation between recurrence, metastasis and survival rate ($p<0.0001$).

Local tumour control

Seven out of 78 patients presented recurrence of the irradiated uveal melanoma. Local tumour control was achieved in 91.0% of patients (71/78). Tumour thickness values, before and after treatment, are shown in table 4. Significant tumour thickness reduction was observed during follow-up ($p<0.0001$).

Eye retention rate

Eight treated eyes were enucleated; four due to tumour recurrence and four due to subsequent ocular complications (recalcitrant pain, neovascular glaucoma and phthisis bulbi). The eye retention rate was 89.7% (70/78 patients).

Visual acuity

Visual acuity data before and after treatment are shown in table 4. A significant absolute reduction was observed during follow-up ($p<0.0001$). Significant correlation was observed between the baseline visual acuity and the absolute variation of visual acuity after GKR ($r = -0.720$, $p<0.0001$) or the relative variation of visual acuity after GKR ($r = -0.274$, $p = 0.015$). A significant relative reduction of visual acuity (-94.4% (-100% to 0%)) was observed during follow-up ($p<0.0001$).

Treatment-related complications

There were few acute ocular complications following our treatment protocol with GKR. The most frequently encountered complications were minor cutaneous bleeding following fixation of the stereotactic frame and subconjunctival haemorrhage due to transconjunctival sutures. Early side effects due to

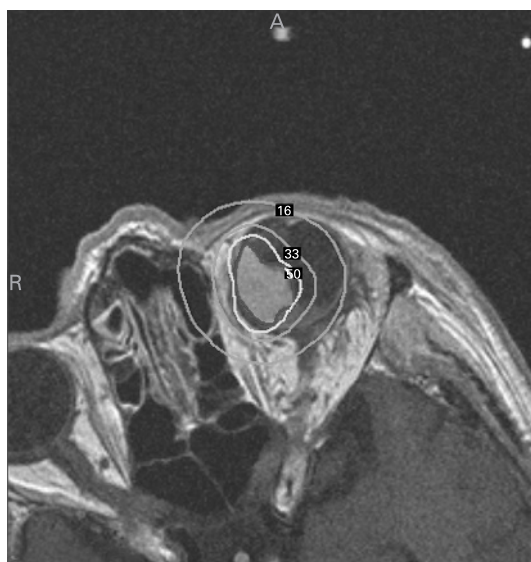


Figure 1 Stereotactic magnetic resonance image showing the treatment plan.

single high-dose radiation were small transient retinal haemorrhages over the surface of the tumour.

Subsequent ocular complications occurred in 63 out of 78 patients, including: exudative retinopathy (33.3%), neovascular glaucoma (18.7%), radiogenic retinopathy (13.5%), vitreous haemorrhages (10.4%), radiogenic optic neuropathy (15.5%), cataract (6.2%) and bulbar phthisis (2.0%).

The occurrence of subsequent ocular proportions during the follow-up period showed a significantly linear trend ($p = 0.0053$) (≤ 12 months: 45%; 12.1–24 months: 71%; 24.1–36 months: 93%; 36.1–48 months: 100%; 48.1–60 months: 78%; >60 months: 94%).

DISCUSSION

In the 1950s, the Swedish neurosurgeon Lars Leksell and the radiobiologist Borje Larsson developed the Gamma Knife Unit for stereotactic radiosurgery of intracranial lesions. GKR is engineered to deliver a single dose of ionising radiation to a small target with steep dose fall-off at the margins. These characteristics are crucial to irradiating intraocular tumours while sparing peritumorous healthy tissue.⁸

In 1987, Rand *et al* showed total regression of anterior chamber melanomas in six rabbit eyes after GKR in a single dose of 60–90 Gy delivered at 90% isodose.¹¹ In 1989, Zambrano *et al* described the first protocol treatment on humans and reported the results of a series of patients with uveal melanoma treated with GKR.¹² Thereafter, many authors reported series of patients with uveal melanoma treated with GKR.^{13–19} Recently, Mueller *et al*, Langmann *et al* and Zehetmayer *et al* described their extensive experience in treating uveal melanoma with GKR.^{9 10 19} The primary radiobiological goal of GKR is to obtain a total radiation effect within the target volume.²⁰ For this reason, immobilisation of the globe and tumour imaging are two of the most important steps in the treatment protocol. Mueller *et al* considered adequate a retrobulbar injection of a long-acting anesthetic agent to obtain complete akinesia during treatment and perform a second MRI to check the post-treatment tumour position.⁹ Zehetmayer built and used a suction attachment device to immobilise the globe,¹⁸ while Langmann *et al* preferred to immobilise the globe using

Table 1 Characteristics at baseline of 78 patients with malignant melanoma

	Median	(Q ₁ –Q ₃)
Follow-up (months)	31.3	(17.6 to 60.6)
Age (years)	64	(58 to 71)
Tumour thickness (mm)	6.1	(4.7 to 8.8)
Visual acuity (Snellen)	0.3	(0.05 to 0.8)
Gy at tumour margin	35	(35 to 40)
Isodose (%)	50	(50 to 55)
	N	%
Sex		
Male	37	47.4
Female	41	52.6
Tumour location		
Ciliary body	3	3.8
Midperiphery	30	38.5
Peripapillary	14	17.9
Posterior pole	31	39.7
Laterality		
Right	47	60.3
Left	31	39.7
Gy at tumour margin		
35	47	60.3
40	23	29.5
50	8	10.3

retrobulbar anesthesia and suture extraocular muscles.¹⁰ In our protocol we used a retrobulbar injection of long-acting agents to obtain complete akinesia. We found that this goal can be achieved through the suture of only two extraocular muscles to immobilise and orientate the globe.

Logani reported that ocular melanoma cell lines are relatively radioresistant in vitro especially at lower doses, while they may be more responsive to a single high dose delivered by stereotactic radiosurgery or high-dose brachytherapy.²¹ In their first reports, many authors used high doses to be sure of destroying tumour cells ranging from 90 to 50 Gy at tumour margins.^{13–19} Then, in order to reduce secondary radiation-induced side effects, the irradiation dose was gradually reduced over the years. Langmann showed that if the dose is reduced from 50 to 40 Gy, there is no influence on local tumour control rate, and this can minimise subsequent side effects.²² Mueller *et al* reported that current treatment is performed using 25 Gy at 50% isodose.⁹ Zehetmayer *et al* showed that a total dose of 45–70 Gy delivered in one to three fractions was effective in

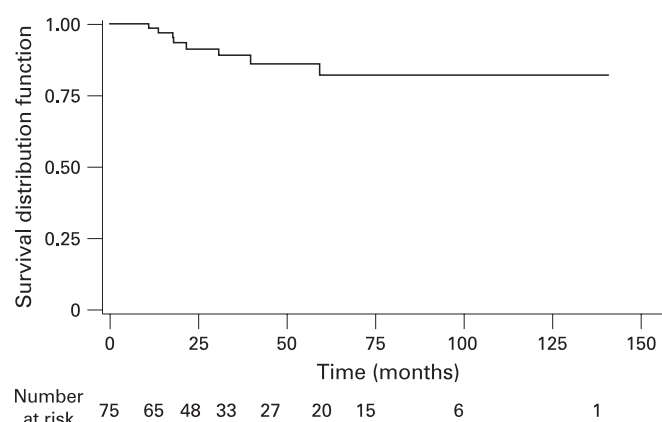


Figure 2 Kaplan–Meier curve of overall survival.

Table 2 Survival probability (%) according to baseline factors (N = 75)

Factor	Category	N	1-year survival	3-year survival	5-year survival	p Value
Age (years)	≤ 64 (median)	39	100	97	86	0.132
	> 64	36	96	77	77	
Sex	Male	36	97	89	89	0.646
	Female	39	100	89	77	
Gy at tumour margin	35	47	100	94	73	0.451
	40	21	95	88	88	
	50	7	100	67	67	
Tumour location	Ciliary body	3	100	67	67	0.390
	Midperiphery	29	96	96	96	
	Peripapillary	14	100	71	71	
	Posterior pole	29	100	95	80	
Laterality	Right	45	100	82	77	0.203
	Left	30	96	96	88	
Tumour thickness	3.1 to 5.0 mm	27	96	86	79	0.552
	5.1 to 10 mm	40	100	89	84	
	>10 mm	8	100	100	100	

reducing radiation induced side effects.¹⁹ In our sample, we reduced the therapeutic dose over the years. The reduction of the dose from 50 to 35 Gy at the tumour margin was not statistically related to survival probability (table 2).

Local tumour control was achieved in 91% of patients. Similar data were presented by Langmann *et al*, Mueller *et al* and Zehetmayer *et al* using GKR.^{9 10 19} In our study we found that local tumour recurrence was strongly related to survival probability ($p < 0.0001$). Seven patients developed local recurrence; five of these patients died from metastatic disease, while the other two patients are alive and metastasis-free.

The therapy for melanoma recurrence was enucleation in four patients and transpupillary thermotherapy in one patient, while two patients were not treated due to concomitant metastatic disease. The number of tumour recurrences evaluated in this study is too small to draw any conclusions concerning clinical prognostic values. In our study, the 3-year survival probability was 88.8 (11.2)% and 5-year survival probability was 81.9 (18.1)%. These data were similar to the survival probability found by Rennie *et al*, which compared GKR and enucleation for uveal melanomas.²³ Moreover, our data on 3- and 5-year cumulative survival rates are comparable with those reported by others on melanomas treated with heavy-particle fraction radiotherapy, brachytherapy, GKR and enucleation.^{4-7 9-19} In our sample there were no statistical correlations between survival probability and age, sex, laterality, Gy at tumour margin,

tumour location and baseline tumour thickness (table 2). The eye retention rate was 89.7%. Four eyes were enucleated because of tumour recurrence and four due to subsequent treatment-related complications (painful eye, neovascular glaucoma and phthisis bulbi). If we consider only the eyes enucleated because of GKR side effects, the eye retention rate increases to 94.8%. These data were similar to the data reported by other authors that used different treatment methods.⁴⁻⁷ As previously demonstrated by other authors, we found that visual acuity results are strictly related to different factors such as vision at baseline, tumour location and size, irradiation volume and subsequent occurrence of radiation side-effects. Our visual acuity data at the end of follow-up showed a statistically significant reduction after treatment (table 4) and a significant correlation between visual acuity loss and duration of follow-up.

As in other reports,²⁴ the most frequent acute complications that we encountered after GKR were subconjunctival haemorrhages due to transconjunctival sutures and small transient retinal haemorrhages over the surface of the tumour. The late appearance of radiation side effects was secondary to the larger tumour size, peripapillary location and longer follow-up. The most frequently encountered complications were: radiogenic retinopathy, neovascular glaucoma and vitreous haemorrhages. Our data showed a linear correlation between the occurrence of radiation ocular side effects and duration of follow-up.

Table 3 Survival probability (%) according to follow-up information (N = 75)

Factor	Category	N	1-year survival	3-year survival	5-year survival	p Value
Metastasis	Yes	9	89	42	14	<0.0001
	No	66	100	98	98	
Recurrence	Yes	7	86	57	19	<0.0001
	No	68	100	93	93	

Table 4 Median (interquartile range) variation of tumour thickness and visual acuity (N = 78)

	Before treatment median (Q ₁ –Q ₃)	After treatment (end of follow-up) median (Q ₁ –Q ₃)	Change* median (Q ₁ –Q ₃)	p Value
Tumour thickness (mm)	6.1 (4.7 to 8.8)	3.9 (2.7 to 5.7)	–1.96 (–3.5/–1.0)	<0.0001
Visual acuity (Snellen)	0.3 (0.05 to 0.8)	0 (0 to 0.05)	–0.2 (–0.6/0)	<0.0001

*Change was calculated between the baseline (before treatment) and the end of follow-up values (whenever it occurred).

CONCLUSIONS

From the present study, we can conclude that the results regarding survival, local tumour control and eye retention rate are encouraging. However, the occurrence of radiation-induced side-effects and visual acuity loss with this technique need to be improved with further reduction in therapeutical dose. Although we recognise that data regarding follow-up are relatively short, and the number of studied patients is limited, GKR should be considered as an alternative treatment for uveal melanomas to enucleation in particular when other conservative treatments are not available or suitable.

Acknowledgements: The authors thank M John for the English-language editing of the manuscript.

Competing interests: None.

Ethics approval: Ethics approval was provided by HSR Ethics Committee.

Patient consent: Obtained.

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